Working with Mock Healthcare Dataset

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AI in Healthcare (ITAI 4375)

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Healthcare Patient Data Analysis: Statistical Calculations

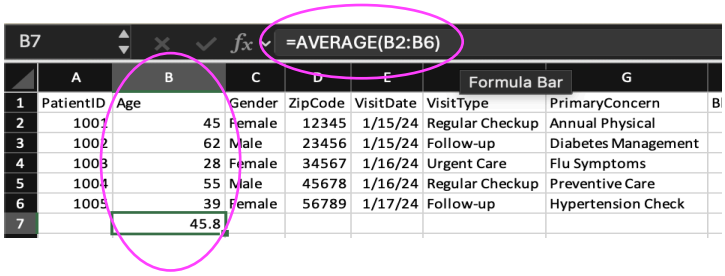
In this assignment, I am working with a mock healthcare dataset from clinic visits. The purpose of this activity is to recognize common sources of healthcare data, describe key healthcare databases, and effectively communicate medical terminology related to AI. By adding the data into a clean excel spreadsheet and calculating the average patient age and average heart rate, I am practicing how to work with patient information and understand what the numbers tell us about healthcare data.

Dataset Overview

The dataset I analyzed comprises five patient records from an outpatient clinic. Each record includes demographic details such as patient ID, age, and gender, along with clinical measurements including cardiovascular readings (blood pressure and heart rate), weight metrics (body mass index), tobacco use history, information about ongoing health conditions, prescription counts, and whether the patient was scheduled for a follow-up appointment.. These five patient visits occurred between January 15 and January 17, 2024.

Calculation 1: Average Patient Age

I determined the mean age of the patient population by extracting all age data and applied a simple mathematical function. The average is computed by going into a cell below the age data and using a spreadsheet formula (=AVERAGE) that automatically sums all the ages and divides by the total number of patients. I specifically typed =AVERAGE(B2:B6), which calculated the average age across all five patients.



	A	B	C	D	E		G
1	PatientID	Age	Gender	ZipCode	VisitDate	VisitType	PrimaryConcern
2	1001	45	Female	12345	1/15/24	Regular Checkup	Annual Physical
3	1002	62	Male	23456	1/15/24	Follow-up	Diabetes Management
4	1003	28	Female	34567	1/16/24	Urgent Care	Flu Symptoms
5	1004	55	Male	45678	1/16/24	Regular Checkup	Preventive Care
6	1005	39	Female	56789	1/17/24	Follow-up	Hypertension Check
7		45.8					

Patient Ages from the Dataset:

- Patient 1001: 45 years
- Patient 1002: 62 years
- Patient 1003: 28 years
- Patient 1004: 55 years
- Patient 1005: 39 years

Average Patient Age: 45.8 years

Calculation 2: Average Heart Rate

I determined the mean heart rate of the patient population by extracting all heart rate data and applied a simple mathematical function. The average is computed by going into a cell below the heart rate data and using a spreadsheet formula (=AVERAGE) that automatically sums all the heart rates and divides by the total number of patients. specifically typed =AVERAGE(J2:J6), which calculated the average heart rate across all five patient measurements.

J7 =AVERAGE(J2:J6)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PatientID	Age	Gender	ZipCode	VisitDate	VisitType	PrimaryConcern	BloodPressureSystolic	BloodPressureDiastolic	HeartRate	BMI	Smoker	ChronicCondition	MedicationsPres
2	1001	45	Female	12345	1/15/24	Regular Checkup	Annual Physical	120	80	72	24.5	No	No	
3	1002	62	Male	23456	1/15/24	Follow-up	Diabetes Management	138	88	78	29.2	Former	Yes	
4	1003	28	Female	34567	1/16/24	Urgent Care	Flu Symptoms	118	76	88	22.1	No	No	
5	1004	55	Male	45678	1/16/24	Regular Checkup	Preventive Care	142	92	76	31.5	Yes	Yes	
6	1005	39	Female	56789	1/17/24	Follow-up	Hypertension Check	128	84	70	26.8	No	Yes	
7		45.8								76.8				

Heart Rate Measurements from the Dataset:

- Patient 1001: 72 beats per minute
- Patient 1002: 78 beats per minute
- Patient 1003: 88 beats per minute
- Patient 1004: 76 beats per minute
- Patient 1005: 70 beats per minute

Average Heart Rate: 76.8 beats per minute

Conclusion

I have successfully calculated two statistical measures from the mock healthcare dataset. The average patient age is 45.8 years, representing a diverse population spanning early adulthood through older age. The average heart rate of 76.8 beats per minute falls within the normal range of 60-100 beats per minute for adults (Mayo Clinic, 2024).

References

Mayo Clinic. (2024). *Heart rate: What's normal?* Retrieved from https://www.mayoclinic.org/healthy-lifestyle/fitness/expert-answers/heart-rate/faq-20057979